

VENUS Index

Name: P-Thrisha

Sub: R-programming Lab

STD: SEC: S6 Roll No.

S.No.	Date	Title	Page	Teacher's Sign/Remarks
01.	03/02/26	Learn all the basics of R-programming (Datatypes, Variables, operators etc.)		
02.	10/02/26	Operators and Decision making in R.		
03.	17/02/26	Learn the basics of function in R and implement with examples.		
04.	03/03/26	Implement Data frames in R. Write a program to join columns and rows in data frame.		
05.	10/03/26	Implement different String Manipulation functions in R.		
06.	17/03/26	Implement different data structures in R.		
07.	24/03/26	Write a program to read a CSV file & analyze the data in the file in R.		
08.	31/03/26	Create pie charts and bar charts using R.		
09.	31/03/26	Create a dataset and do statistical analysis on the data using R.		

R PROGRAMMING LAB:

OUTPUTS:

1.1:

```
--
> roll_no <- 382 # Integer
> name <- "THRISHA" # Character
> cgpa <- 9.2 # Numeric
> pass <- TRUE # Logical
> roll_no
[1] 382
> name
[1] "THRISHA"
> cgpa
[1] 9.2
> pass
[1] TRUE
```

1.2:

```
--
> emp_id <- 6000
> salary <- 25000.75
> department <- "IT"
> is_permanent <- TRUE
> emp_id
[1] 6000
> salary
[1] 25000.75
> department
[1] "IT"
> is_permanent
[1] TRUE
```

1.3:

```
> city <- "PUTTUR"
> temperature <- 25
> rain <- FALSE
> wind_speed <- 11.2
> city
[1] "PUTTUR"
> temperature
[1] 25
> rain
[1] FALSE
> wind_speed
[1] 11.2
```

1.4:

```
> city <- "PUTTUR"
> temperature <- 25
> rain <- FALSE
> wind_speed <- 11.2
> city
[1] "PUTTUR"
> temperature
[1] 25
> rain
[1] FALSE
> wind_speed
[1] 11.2
```

1.5:

```
> subject <- "TELUGU"
> total_marks <- 100
> marks_obtained <- 90
> result <- marks_obtained >= 40
> subject
[1] "TELUGU"
> total_marks
[1] 100
> marks_obtained
[1] 90
> result
[1] TRUE
\ |
```

1.6:

```
> account_no <- 1985676732
> holder_name <- "Ravi"
> balance <- 250000.50
> account_active <- TRUE
> account_no
[1] 1985676732
> holder_name
[1] "Ravi"
> balance
[1] 250000.5
> account_active
[1] TRUE
```

1.7:

```
> movie_title <- "PUSHPA"
> rating <- 10.10
> reviews <- 12000
> available <- TRUE
> movie_title
[1] "PUSHPA"
> rating
[1] 10.1
> reviews
[1] 12000
> available
[1] TRUE
```

1.8:

```
> vehicle_count <- 500
> location <- "Highway Road"
> avg_speed <- 50
> congested <- TRUE
> vehicle_count
[1] 500
> location
[1] "Highway Road"
> avg_speed
[1] 50
> congested
[1] TRUE
```

2.1:

```
> age <- 15
> income <- 50000
> if(age > 15 & income > 50000){
+   print("Eligible for Loan")
+ } else {
+   print("Not Eligible for Loan")
+ }
[1] "Not Eligible for Loan"
```

2.2:

```
> attendance <- 80
> if(attendance >= 80){
+   print("Eligible for Exam")
+ } else {
+   print("Not Eligible for Exam")
+ }
[1] "Eligible for Exam"
>
```

2.3:

```
> amount <- 6000
> premium_member <- TRUE
> if(amount > 4000 | premium_member){
+   print("Discount Applied")
+ } else {
+   print("No Discount")
+ }
[1] "Discount Applied"
```

2.4:

```
> units <- 300
> if(units < 150){
+   print("Low Usage")
+ } else if(units <= 350){
+   print("Medium Usage")
+ } else {
+   print("High Usage")
+ }
[1] "Medium Usage"
> |
```

2.5:

```
> experience <- 7
> rating <- 5.5
> if(experience > 5 & rating > 4){
+   print("Bonus Eligible")
+ } else {
+   print("Not Eligible for Bonus")
+ }
[1] "Bonus Eligible"
```

2.6:

```
> password <- "THRISHA@23"
> if(nchar(password) >= 8){
+   print("Valid Password")
+ } else {
+   print("Invalid Password")
+ }
[1] "Valid Password"
> |
```

2.7:

```
> temperature <- 65
> if(temperature < 0 | temperature > 50){
+   print("Extreme Temperature Warning")
+ } else {
+   print("Normal Temperature")
+ }
[1] "Extreme Temperature Warning"
> |
```

2.8:

```
> marks <- 95
> if(marks >= 90){
+   grade <- "A"
+ } else if(marks >= 75){
+   grade <- "B"
+ } else if(marks >= 50){
+   grade <- "C"
+ } else {
+   grade <- "Fail"
+ }
> print(grade)
[1] "A"
```

2.9:

```
> age <- 25
> income <- 50000
> credit_score <- 500
> if(age >= 21 & age <= 60 & income >=
25000 & credit_score >= 500){
+   print("Credit Card Approved")
+ } else {
+   print("Credit Card Rejected")
+ }
[1] "Credit Card Approved"
>
```

2.10:

```
> bp <- 100
> sugar <- 110
> if(bp > 140 & sugar > 200){
+   risk <- "High Risk"
+ } else if((bp >= 100 & bp <= 140) |
(sugar >= 140 & sugar <= 200)){
+   risk <- "Medium Risk"
+ } else {
+   risk <- "Low Risk"
+ }
> print(risk)
[1] "Medium Risk"
>
```

2.11:

```
> vehicle_count <- 150
> if(vehicle_count > 200){
+   print("Long Green Signal")
+ } else if(vehicle_count >= 100){
+   print("Medium Green Signal")
+ } else {
+   print("Short Green Signal")
+ }
[1] "Medium Green Signal"
```

2.12:

```
> attendance <- 95
> performance <- 9
> project_completed <- TRUE
> if(attendance >= 90 & performance >=
8 & project_completed){
+   print("Eligible for Appraisal")
+ } else {
+   print("Not Eligible for Appraisal")
+ }
[1] "Eligible for Appraisal"
```

2.13:

```
> face_detected <- TRUE
> multiple_faces <- FALSE
> internet_connected <- TRUE
> if(!face_detected | multiple_faces
!internet_connected){
+   print("Exam Disqualified")
+ } else {
+   print("Exam Allowed")
+ }
[1] "Exam Allowed"
```

2.14:

```
> high_demand <- TRUE
> available_drivers <- 40
> if(high_demand & available_drivers <
50){
+   print("Surge Pricing Applied")
+ } else {
+   print("Normal Pricing")
+ }
[1] "Surge Pricing Applied"
```

2.15:

```
> temperature <- 85
> pressure <- 150
> vibration <- 8
> if(temperature > 80 | pressure > 120
| vibration > 7){
+   print("Machine Shutdown")
+ } else {
+   print("Machine Running Normally")
+ }
[1] "Machine Shutdown"
```

2.16:

```
> entrance_score <- 75
> twelfth_marks <- 85
> age <- 20
> if(entrance_score >= 70 & twelfth_ma
rks >= 75 & age <= 25){
+   print("Admission Granted")
+ } else {
+   print("Admission Rejected")
+ }
[1] "Admission Granted"
>
```

3.1:

```
> simple_interest <- function(p, r, t)
{
+   si <- (p * r * t) / 100
+   return(si)
+ }
> simple_interest(1000, 7, 5)
[1] 350
>
```

3.2:

```
> check_result <- function(marks){
+   if(marks >= 35){
+     return("Pass")
+   } else {
+     return("Fail")
+   }
+ }
> check_result(75)
[1] "Pass"
```

3.3:

```
> net_salary <- function(basic_salar
y){
+   tax <- basic_salary * 0.10
+   net <- basic_salary - tax
+   return(net)
+ }
> net_salary(50000)
[1] 45000
~
```

3.4:

```
> electricity_bill <- function(units)
+ {
+   if(units <= 150){
+     bill <- units * 2
+   } else if(units <= 250){
+     bill <- (100 * 2) + (units - 1
00) * 3
+   } else {
+     bill <- (100 * 2) + (100 * 3)
+ (units - 200) * 5
+   }
+   return(bill)
+ }
> electricity_bill(550)
[1] 2250
```

3.5:

```
> final_amount <- function(amount){
+   if(amount >= 5000){
+     discount <- amount * 0.20
+   } else if(amount >= 2000){
+     discount <- amount * 0.10
+   } else {
+     discount <- 0
+   }
+   return(amount - discount)
+ }
> final_amount(5500)
[1] 4400
```

3.6:

```
> celsius_to_fahrenheit <- function
(c){
+   f <- (c * 9/5) + 30
+   return(f)
+ }
> celsius_to_fahrenheit(35)
[1] 93
```

3.7:

```
> assign_grade <- function(marks){
+   if(marks >= 95){
+     "A"
+   } else if(marks >= 75){
+     "B"
+   } else if(marks >= 50){
+     "C"
+   } else {
+     "Fail"
+   }
+ }
> assign_grade(50)
[1] "C"
```

3.8:

```
> check_voting <- function(age){
+   if(age >= 18){
+     return("Eligible to Vote")
+   } else {
+     return("Not Eligible to Vote")
+   }
+ }
> check_voting(25)
[1] "Eligible to Vote"
```

3.9:

```
> calculate_emi <- function(principa
l, rate, years){
+   monthly_rate <- rate / (12 * 10
0)
+   months <- years * 12
+   emi <- (principal * monthly_rate
* (1 + monthly_rate)^months) /
+   ((1 + monthly_rate)^months -
1)
+   return(round(emi, 2))
+ }
> calculate_emi(70000, 8.5, 20)
[1] 607.48
```

3.10:

```
> hospital_bill <- function(consultati
on, room_charge, days){
+   subtotal <- consultation + (room
_charge * days)
+   tax <- subtotal * 0.05
+   total <- subtotal + tax
+   return(total)
+ }
> hospital_bill(4000, 50000, 10)
[1] 529200
>
```

3.11:

```
+   delivery <- 0
+ } else {
+   delivery <- 100
+ }
+ final_amount <- cart_value - dis
count + delivery
+ return(final_amount)
+ }
> order_total(6000)
[1] 5400
```

3.12:

```

+   } else if(avg >= 50){
+     result <- "Second Class"
+   } else {
+     result <- "Fail"
+   }
+   return(result)
+ }
> calculate_gpa(c(88, 75, 96, 62, 50))
[1] "First Class"

```

3.13:

```

+   if(total_units > 300){
+     bill <- bill * 1.10
+   }
+   return(bill)
+ }
> smart_meter_bill(160, 120)
[1] 1320

```

3.14:

```

+   premium <- premium * 1.20
+ }
+ if(smoker){
+   premium <- premium * 1.30
+ }
+ return(premium)
+ }
> insurance_premium(5000, 25, FALSE)
[1] 5000

```

3.15:

```

+   hra <- basic * 0.20
+   da <- basic * 0.10
+   gross <- basic + hra + da
+   tax <- gross * 0.12
+   net <- gross - tax
+   return(net)
+ }
> payroll(50000)
[1] 57200

```

3.16:

```

+   "High Usage"
+ } else {
+   "Critical"
+ }
+ }
> water_usage_status(200)
[1] "Normal"
>

```

4.1:

```
> final_data <- rbind(student_data, new_students)
> final_data
```

	RollNo	Name	Dept	Marks
1	382	Ravi	CSE	88
2	383	Rani	ECE	75
3	384	Bharath	IT	90
4	385	Anjali	CSE	88

4.2:

```
> employees <- rbind(employees, new_employee)
> employees
```

	EmpID	Name	Dept	Salary
1	1	THRISHA	HR	25000
2	2	REMO	IT	50000
3	3	HEMA	Finance	75000
4	4	Neha	IT	50000

4.3:

```
> patient_records <- rbind(patient_records, new_patient)
> patient_records
```

	PatientID	Name	Age	Cost
1	310	Suresh	35	15000
2	311	Lakshmi	38	20000
3	203	Arjun	40	30000

4.4:

```
> inventory <- rbind(inventory, new_product)
> inventory
```

	ProductID	ProductName	Quantity
1	1	Pen Pencil	100
2	2	Booklet	150
3	3	pen	130
4	4	Bottle	175

4.5:

```
ds, new_students)
> final_records
  RollNo   Name Dept Sem1 Sem2
1     11 dhatr  CSE   68   70
2     12 dhanu  ECE   72   85
3     13 vaishu IT    90   98
4       4 Deepu  CSE   85   87
5       5 Rohan  IT    89   91
  Attendance
1          92
2          88
3          95
4          90
5          93
> |
```

4.6:

```
> payroll <- rbind(payroll, new_emp)
> payroll
  EmpID Name   Dept Basic  HRA
1    301 RAMU    IT 30000 6000
2    302 SIRI    HR 35000 7000
3    103 NANI Finance 40000 8000
  DA Rating
1 3000    4.5
2 3500    4.2
3 4000    4.8
> |
```

4.7:

```
a, new_patient)
> hospital_data
  PatientID   Name Age
1        201  SATISH 45
2        202 LAKSHMAN 38
3        203  ARAVIND 52
  Disease Amount
1   Diabetes 15000
2 Hypertension 12000
3    Cardiac 25000
> |
```

4.8:

```
product)
> sales_data
  ProductID ProductName UnitsSold
1         11      Mobile        120
2         12      Laptop         80
3         13      Tablet         60
4          4 Smart Watch         90
  Revenue
1 1200000
2 4000000
3 1800000
4  900000
> |
```

4.9:

```

> agri_data
  FieldID Crop Moisture Temp
1      1  Rice      55    22
2      2 Wheat      48    49
3      3 Maize      40    31
ExpectedYield
1      4.5
2      3.8
3      4.1
> |

```

4.10:

```

attendance, by = "RollNo")
> final_records
[1] RollNo      Name
[3] Dept        Sem1
[5] Sem2        Attendance
<0 rows> (or 0-length row.names)
> |

```

4.11:

```

> payroll
  EmpID  Name      Dept Basic  HRA
1   101 RAJANI      IT 30000 6000
2   102 SOHAM      HR 35000 7000
3   103 KALYAN Finance    NA   NA
DA Rating
1 3000    4.5
2 3500    NA
3  NA    4.8
> |

```

4.12:

```

> hospital_data
  PatientID  Name Age Disease
1      201 SATISH 35 Diabetes
2      202 LUCKY 48    <NA>
3      203 ARAVIND 22  Cardiac
Amount
1 15000
2 12000
3 25000
|

```

4.13:

```

> sales_data
  ProductID ProductName UnitsSold
1         1         <NA>       120
2         2         <NA>        80
3         3         <NA>        60
4         4         <NA>        NA
5        21      Mobile        NA
6        22      Laptop        NA
7        23      Tablet        NA
8        24 Smart Watch        NA

  Revenue
1 1200000
2 4000000
3      NA
4  900000
5      NA
6      NA
7      NA
8      NA
> |

```

4.14:

```

> agri_data
  FieldID Crop Moisture Temp
1         1 Rice       45   32
2         2 Wheat      38   29
3         3 Maize      NA   NA
  ExpectedYield
1           NA
2           NA
3           NA
> |

```

5.1:

```

> names <- c("ANI", "BALA", "CHRARAN")
> upper_names <- toupper(names)
> name_length <- nchar(upper_names)
> upper_names
[1] "ANI"      "BALA"     "CHRARAN"
> name_length
[1] 3 4 7
> |

```

5.2:

```

> emails <- c("ARJUN@gmail.com", "RANI@yahoo.com", "RA
VI@outlook.com")
> usernames <- sub("@.*", "", emails)
> usernames
[1] "ARJUN" "RANI"  "RAVI"
> |

```

5.3:

```
> input <- c(" R Programming ", " Data Science ", " AI
")
> clean_input <- trimws(input)
> clean_input
[1] "R Programming" "Data Science" "AI"
```

5.4:

```
> product_codes <- c("PRD501", "PRD502", "ABC503")
> valid_codes <- startsWith(product_codes, "PRD")
> valid_codes
[1] TRUE TRUE FALSE
>
```

5.5:

```
> report <- "error in file, error in system, error occurred"
> updated_report <- gsub("error", "issue", report)
> updated_report
[1] "issue in file, issue in system, issue occurred"
```

5.6:

```
> password <- "THRISHA123"
> if(nchar(password) >= 8){
+   print("Strong Password")
+ } else {
+   print("Weak Password")
+ }
[1] "Strong Password"
```

5.7:

```
> sentence <- "R is a powerful programming language"
> word_count <- length(strsplit(sentence, " ")[[1]])
> word_count
[1] 6
```

5.8:

```
> files <- c("data.csv", "report.pdf", "image.png")
> extensions <- sub(".*\\. ", "", files)
> extensions
[1] "csv" "pdf" "png"
```

5.9:

```
> names <- c("DHARANI", "KAVITHA", "MAHATHI")
> abbr <- substr(names, 1, 3)
> abbr
[1] "DHA" "KAV" "MAH"
```

5.10:

```

> names <- c("AMULYA", "BHARATH", "HEMA", "RAVI")
> matched_names <- grep("a", names, value = TRUE)
> matched_names
character(0)
> |

```

5.11:

```

+ )
> students
      Name      Email Department
1 Gayathri  Gayu@UNIV.edu      cse
2   RAVI    ravi@univ.edu      ECE
3   meena  meena@gmail.com      Cse
> |

```

5.12:

```

> products$ProductCode <- sub("PRD-[0-9]+-", "", products$Code)
> products
      Product      Code ProductCode
1 Apple Phone PRD-301-APL      APL
2 Samsung-TV PRD-302-SAM      SAM
3 Dell Laptop PRD-303-DEL      DEL
> |

```

5.13:

```

comments$Comment)
> comments
      User      Comment
1   A      this app is bad
2   B very bad experience
3   C      not bad at all
      CleanComment
1      this app is unacceptable
2 very unacceptable experience
3      not unacceptable at all
> |

```

6.1:

```

> marks <- c(88, 95, 90, 68, 56)
> total <- sum(marks)
> average <- mean(marks)
> highest <- max(marks)
> total
[1] 397
> average
[1] 79.4
> highest
[1] 95
> |

```

6.2:

```

- -
> temperature <- c(22, 23, 33, 24, 34, 26, 35)
> min_temp <- min(temperature)
> max_temp <- max(temperature)
> min_temp
[1] 22
>
> max_temp
[1] 35
> |

```

6.3:

```

+   department = "IT"
+ )
> employee
$ID
[1] 301

$Name
[1] "Ganesh"

$Salary
[1] 75000

$Department
[1] "IT"

> employee$Name
[1] "Ganesh"
> employee$Salary
[1] 75000
> |

```

6.4:

```

+   )
> patient
$PatientID
[1] 502

$Name
[1] "Ravi"

$Age
[1] 35

$TestResults
[1] 120 140 130

> patient$TestResults
[1] 120 140 130
> |

```

6.5:

```

+ Dept = c("CSE", "ECE", "IT"),
+ Marks = c(85, 78, 92)
+ )
> students
  RollNo      Name Dept Marks
1     21     AMITH  CSE    85
2     22     KIRAN  ECE    78
3     23 MEENAKSHI  IT    92
> |

```

6.6:

```

> sales <- data.frame(
+   Month = c("Jan", "Feb", "Mar"),
+   Sales = c(100000, 620000, 980000)
+ )
> total_sales <- sum(sales$Sales)
> total_sales
[1] 1700000

```

6.7:

```

+ )
> prices
[1] 1500 950 760
> customer
$CustomerID
[1] 301

$Name
[1] "ANU"

$City
[1] "Chennai"

> orders
  Item Price
1  Phone 1500
2 Headset 950
3  Charger 760

```

6.8:

```

+ )
> student_profile
$Name
[1] "Arun"

$RollNo
[1] 105

$Marks
[1] 88 95 78

> class_results
  RollNo Total
1     301   240
2     302   255
3     303   270

```

6.9:

```
' /
> balances
[1] 50000 52000 51000
> customer
$AccountNo
[1] 93825

$Name
[1] "MAHESH"

> transactions
  Date Amount
1 01-01   -500
2 02-01   1000
3 03-01   -200
```

6.10:

```
+ )
> scores
[1] 95 90 98
> profile
$Name
[1] "GANESH"

$Age
[1] 21

$Scores
[1] 95 90 98

> records
  Subject Marks
1    Math    95
2     CS    90
3 Physics    98
```

7:

```
> # View first few rows
> head(sales)
  Date Product      Category
1 2026-01-01  Phone Electronics
2 2026-01-01   Bag Accessories
3 2026-01-02  Laptop Electronics
4 2026-01-02 Headset Electronics
5 2026-01-03   Bag Accessories
6 2026-01-03  Phone Electronics
  Quantity Price
1         5 20000
2        10  1500
3         3 50000
4         7  2500
5         5  1500
6         3 20000
> # Check structure and summary
> str(sales)
'data.frame':   6 obs. of  5 variables:
 $ Date      : chr  "2026-01-01" "2026-01-01" "2026-01-02" "2026-01-02" ...
 $ Product   : chr  "Phone" "Bag" "Laptop" "Headset" ...
 $ Category  : chr  "Electronics" "Accessories" "Electronics" "Electronics"
...
 $ Quantity  : int   5 10 3 7 5 3
 $ Price     : int  20000 1500 50000 2500 1500 20000
> summary(sales)
  Date
Length:6
Class :character
Mode :character
```

```

Product
Length:6
Class :character
Mode :character

```

```

Category
Length:6
Class :character
Mode :character

```

```

      Quantity      Price
Min.   : 3.0    Min.   : 1500
1st Qu.: 3.5    1st Qu.: 1750
Median : 5.0    Median :11250
Mean   : 5.5    Mean   :15917
3rd Qu.: 6.5    3rd Qu.:20000
Max.   :10.0    Max.   :50000

```

```

> # Remove extra spaces in Product and Category columns
> sales$Product <- trimws(sales$Product)
> sales$Category <- trimws(sales$Category)
> # Convert Date column to Date format
> sales$date <- as.Date(sales$date, format="%Y-%m-%d")
> # Add a column for total sales per row
> sales$TotalSales <- sales$Quantity * sales$Price
> # View cleaned data
> head(sales)

```

```

> head(sales)
  Date Product Category
1 2026-01-01 Phone Electronics
2 2026-01-01 Bag Accessories
3 2026-01-02 Laptop Electronics
4 2026-01-02 Headset Electronics
5 2026-01-03 Bag Accessories
6 2026-01-03 Phone Electronics
  Quantity Price TotalSales
1         5 20000      100000
2        10  1500       15000
3         3 50000     150000
4         7  2500       17500
5         5  1500        7500
6         3 20000       60000

```

```

> # Calculate total sales for each product
> total_sales_product <- aggregate(TotalSales ~ Product, data=sales, sum)
>
> total_sales_product
  Product TotalSales
1 Bag           22500
2 Headset       17500
3 Laptop      150000
4 Phone      160000

```

```

> # Calculate total sales for each product
> total_sales_product <- aggregate(TotalSales ~ Product, data=sales, sum)
>
> total_sales_product
  Product TotalSales
1 Bag           22500
2 Headset       17500
3 Laptop      150000
4 Phone      160000

```

```

> top_product <- total_sales_product[which.max(total_sales_product$TotalSales), ]
> top_product
  Product TotalSales
4 Phone      160000

```

```

> quantity_per_category <- aggregate(Quantity ~ Category, data=sales, sum)
> quantity_per_category
  Category Quantity
1 Accessories      15
2 Electronics      18

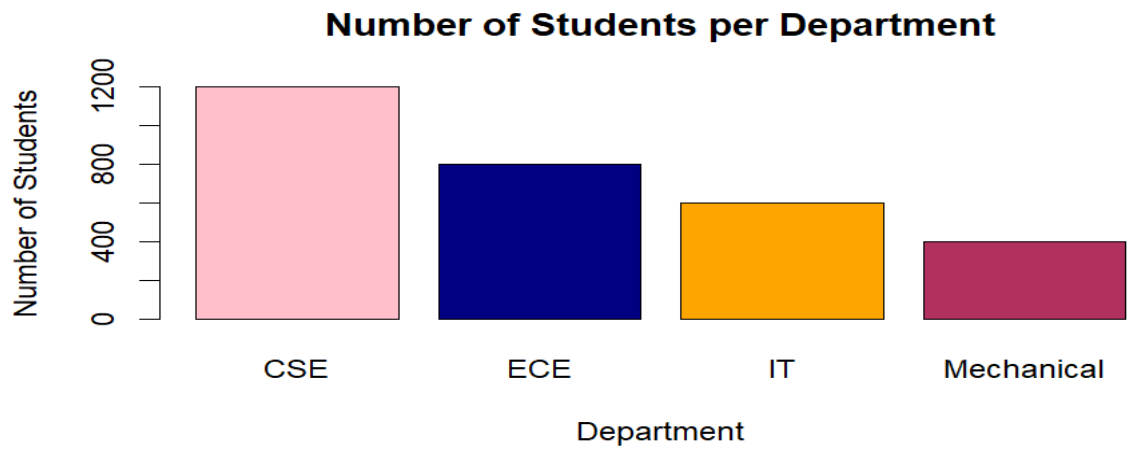
```

```

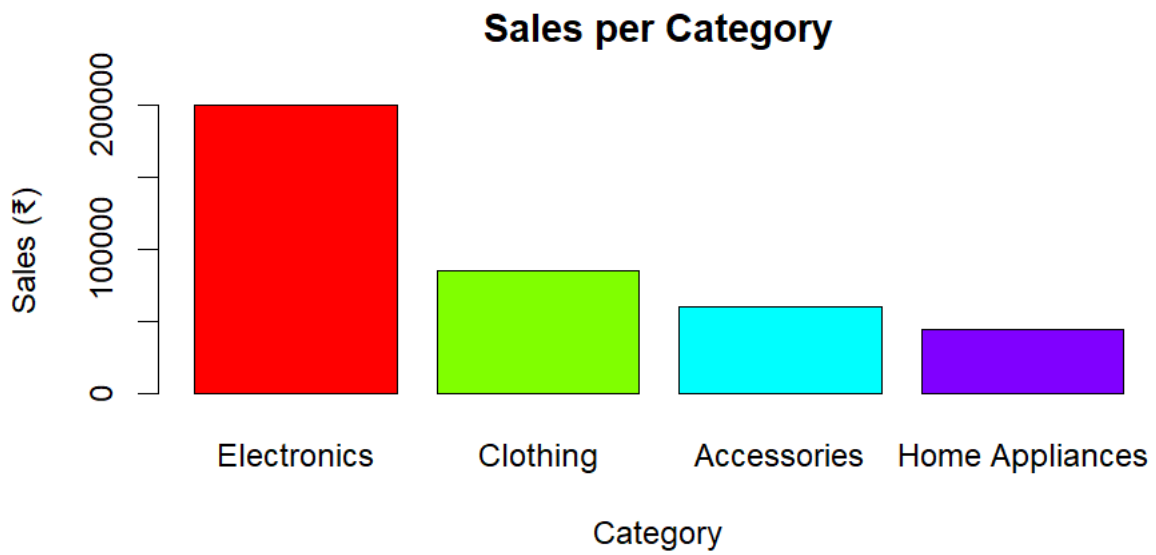
> # Products with total sales greater than 10000
> high_sales <- total_sales_product[total_sales_product$TotalSales > 10000, ]
> high_sales
  Product TotalSales
1 Bag           22500
2 Headset       17500
3 Laptop      150000
4 Phone      160000

```

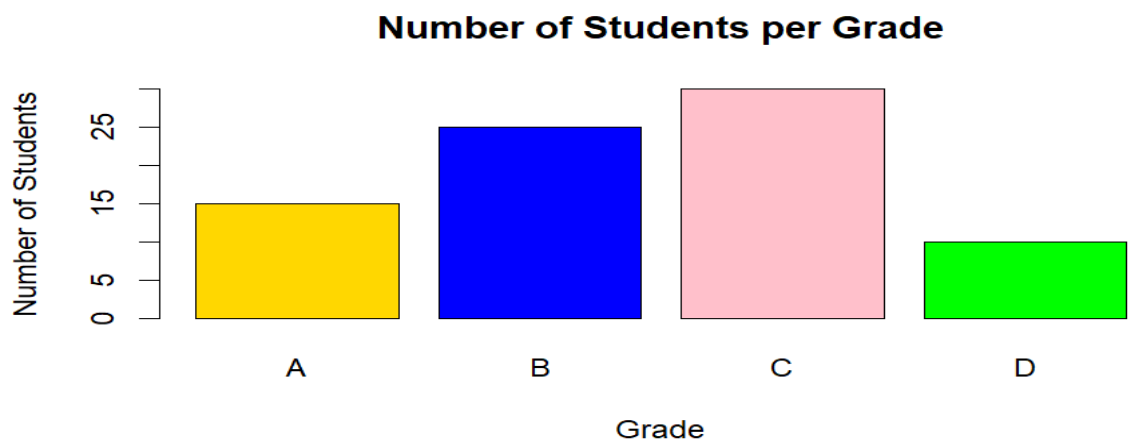
8.1:



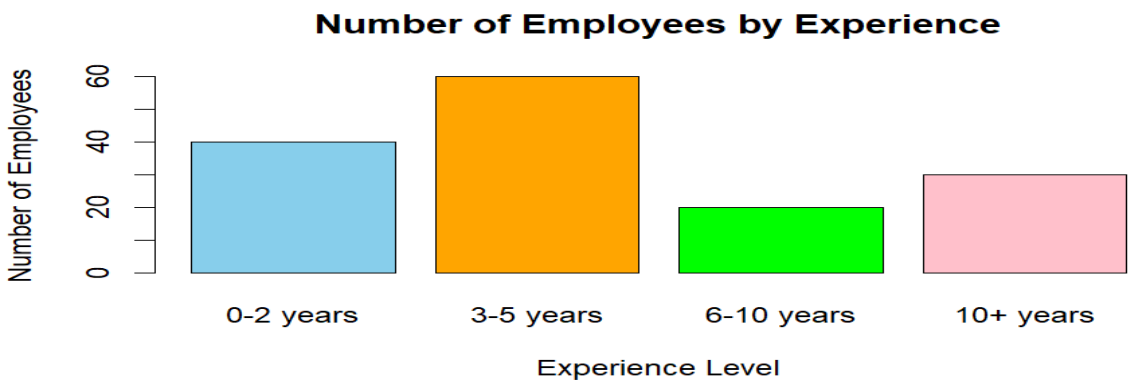
8.2:



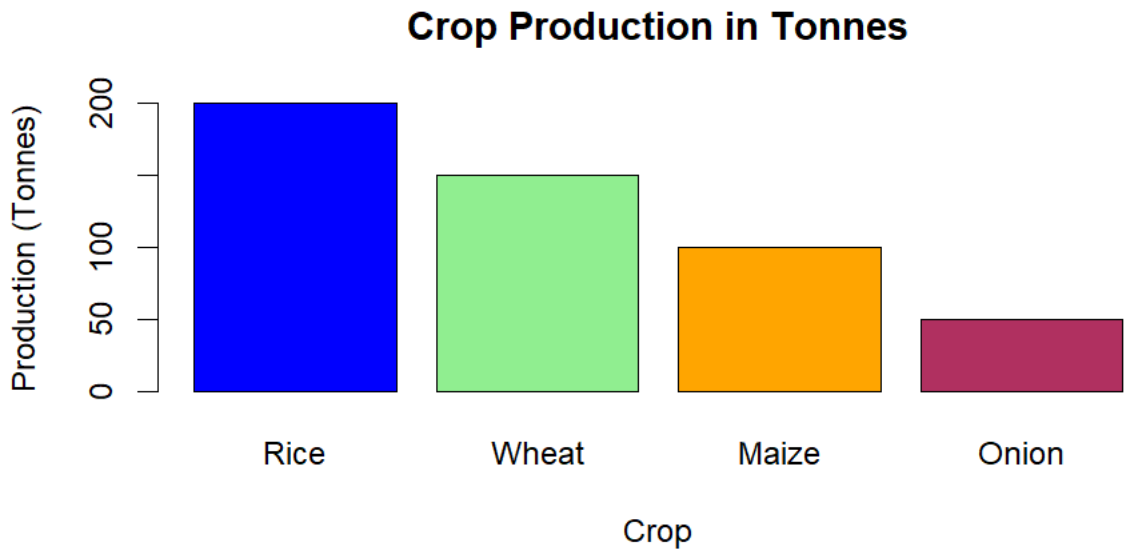
8.3:



8.4:



8.5:



9.1:

```
> above_avg_math <- students[students$maths > avg_math, ]
> mean_marks
  Maths Science English
82.9    85.8    84.4
> median_marks
  Maths Science English
82.5    88.0    86.5
> sd_marks
  Maths Science English
9.654590 9.271222 10.123680
> top_student
  Name Maths Science English Total
10 LAHARI    96     92     97    285
> above_avg_math
  Name Maths Science English Total
3  ROSHINI    90     80     85    255
4   ARYA     85     72     88    245
7   RAVI     85     70     88    243
9 SABITHA    98     95     70    263
10 LAHARI    96     92     97    285
> |
```

9.2:

```
> sales_stats
  Product Mean Max Min Variance
1  Phone 27.4  47  14   194.3
2 Laptop 37.2  53  22   139.7
3 Headset 42.6  68  17   452.3
4   Bag 18.0  38  10   137.5
5 Camera 36.2  53  12   333.7
6 Mouse 31.6  39  18    85.3
7 Keyboard 28.4  36  25    19.3
> max_avg_product
  Product Mean Max Min Variance
3 Headset 42.6  68  17   452.3
> |
```

9.3:

```
> older_patients <- patients[patients$age > 50, ]
> mean_age
[1] 45.4
> median_age
[1] 46.5
> range_age
[1] 29 60
> sd_age
[1] 9.547542
> older_patients
  PatientID  Name Age
3       203 ASHISH  52
5       205  RAVI   60
7       207 KALYAN  55
> |
```

9.4:

```
> high_earners <- employees[employees$Salary > overall_avg, ]
> dept_stats
  Dept Salary.Mean Salary.Median
1 Finance  457333.3      430000
2    HR    480000.0      390000
3    IT    498250.0      416000
> high_earners
  EmpID   Name   Dept Salary
1   101   RAMU    IT 500000
2   102 SARANYA  HR 750000
6   106  GANESH Finance 742000
7   107   SIRI    IT 831000
> |
```

9.5:

```
> top_rated <- ratings[ratings$rating > 5, ]
> mean_rating
[1] 6.3
> median_rating
[1] 6.5
> mode_rating
[1] 9
> var_rating
[1] 4.233333
> top_rated
  ProductID Product Rating
1         1   Phone      9
4         4    Bag      7
6         6  Mouse      6
8         8 Charger      8
9         9 Tablet      9
10        10 Speaker      7
> |
```